For Loops & Range()

A **loop** repeats a block of code. Write the instruction once and let Python repeat it.

1) Why use a for loop?

Without a loop, 100 messages need 100 `print()` lines. A `for` loop repeats code a **known number of times**, or once for every item.

Example :

```
for count in range(3):
    print("Welcome!", count)
```

Output :

```
Welcome! 0
Welcome! 1
Welcome! 2
```

`range(3)` produces 0, 1, 2 → `count` takes each value → indented body runs 3 times

2) range() makes the numbers

Form	Meaning	Produces
`range(5)`	starts at 0, stops before 5	`0, 1, 2, 3, 4`
`range(1, 6)`	starts at 1, stops before 6	`1, 2, 3, 4, 5`
`range(2, 11, 2)`	start, stop, step	`2, 4, 6, 8, 10`

the stop value is excluded: range(1, 6) gives 1 to 5, never 6

3) Multiplication table

Example :

```
table = int(input("Enter a table number: "))

for multiplier in range(1, 11):
    print(f"{table} x {multiplier} = {table * multiplier}")
```

Output :

```
Enter a table number: 7
7 x 1 = 7
7 x 2 = 14
7 x 3 = 21
7 x 4 = 28
7 x 5 = 35
7 x 6 = 42
7 x 7 = 49
7 x 8 = 56
7 x 9 = 63
7 x 10 = 70
```

Quick check

`range(1, 6)` prints **1 2 3 4 5**, and `range(2, 11, 2)` prints **2 4 6 8 10**.

Looping Through Data

A `for` loop can visit each character of a string, each item of a list, or each pair of a dictionary.

1) Strings and lists

Example :

```
word = "PYTHON"
for character in word:
    print(character)
students = ["Asha", "Rahul", "Priya", "Amit"]
for name in students:
    print(name)
```

Output :

```
P
Y
T
H
O
N
Asha
Rahul
Priya
Amit
```

2) Running total

Example :

```
numbers = [10, 20, 30, 40]
total = 0
for number in numbers:
    total = total + number
print(f"The final value is: {total}")
```

Output :

```
The final value is: 100
```

3) Mini project: student marks analyser

Example :

```
students = {"Amit": 75, "Priya": 85, "Rahul": 32,
            "Sneha": 60, "Vijay": 38}
total_marks, passed = 0, 0
for name, marks in students.items():
    total_marks += marks
    if marks >= 40:
        passed += 1
        result = "Pass"
    else:
        result = "Fail"
    print(f"{name}: {marks} - {result}")
print("Total Marks:", total_marks)
print("Average Marks:", total_marks / len(students))
print("Students Passed:", passed)
```

Output :

```
Amit: 75 - Pass
Priya: 85 - Pass
Rahul: 32 - Fail
Sneha: 60 - Pass
Vijay: 38 - Fail
Total Marks: 290
Average Marks: 58.0
Students Passed: 3
```

Quick check

`for name, marks in students.items()` gives **both the key and the value** on every repeat.

While Loops, Break & Continue

A `while` loop repeats **while a condition is True**. Update the value, or it never ends.

1) How a while loop works

start value `countdown = 5` → check `countdown > 0` → run the body then update → False: loop ends

Example :

```
countdown = 5
while countdown > 0:
    print(countdown)
    countdown = countdown - 1
print("Go!")
```

Output :

```
5
4
3
2
1
Go!
```

2) Mini project: ATM withdrawal

Example :

```
balance = 10000
while True:
    print("Balance:", balance)
    amount = int(input("Withdraw (0 to exit): "))
    if amount == 0:
        print("Thank you for using the ATM.")
        break
    elif amount < 0:
        print("Invalid amount! Enter a positive amount.")
    elif amount > balance:
        print("Insufficient balance!")
    else:
        balance -= amount
```

Output :

```
Balance: 10000
Withdraw (0 to exit): 800
Balance: 9200
Withdraw (0 to exit): -5
Invalid amount! Enter a positive amount.
Balance: 9200
Withdraw (0 to exit): 20000
Insufficient balance!
Balance: 9200
Withdraw (0 to exit): 0
Thank you for using the ATM.
```

3) break stops the loop, continue skips one repeat

Example :

```
for number in range(1, 8):
    if number == 4:
        break
    print(number)
```

Output :

```
1
2
3
```

Example :

```
for number in range(1, 6):
    if number == 3:
        continue
    print(number)
```

Output :

```
1
2
4
5
```

Practice Tasks & Revision

Try each task first, then compare. Long outputs are shortened with ...

Task 1 Print a message 5 times.

Solution :

```
for count in range(5):
    print("I am learning Python")
```

```
I am learning Python
I am learning Python
...
```

Task 2 Count from 1 to 10.

Solution :

```
for number in range(1, 11):
    print(number)
```

```
1
2
...
```

Task 3 Even numbers 2 to 20.

Solution :

```
for number in range(2, 21, 2):
    print(number)
```

```
2
4
...
```

Task 4 Multiplication table.

Solution :

```
table = int(input("Enter a number: "))
for i in range(1, 11):
    print(f"{table} x {i} = {table * i}")
```

```
Enter a number: 5
5 x 1 = 5
...
```

Task 5 Sum a list.

Solution :

```
numbers = [5, 10, 15, 20]
total = 0
for number in numbers:
    total = total + number
print("Total:", total)
```

```
Total: 50
```

Task 6 Print each character.

Solution :

```
word = input("Enter a word: ")
for character in word:
    print(character)
```

```
Enter a word: Asha
A
...
```

Task 7 while counter 1 to 5.

Solution :

```
counter = 1
while counter <= 5:
    print(counter)
    counter = counter + 1
```

```
1
2
...
```

Task 8 Stop word.

Solution :

```
word = ""
while word.lower() != "stop":
    word = input("Enter a word (type 'stop'): ")
print("Finished!")
```

```
Enter a word (type 'stop'): hello
Enter a word (type 'stop'): STOP
Finished!
```

Quick revision

Loop	Best beginner use	Example
for	known number of repeats, or every item	for number in range(5):
while	repeat while a condition stays true	while counter <= 5:
break	stop the loop immediately	stop when a target is found
continue	skip one repeat	skip a particular number